Fire Safety Journal 162 (2026) 104724

Contents lists available at ScienceDirect

Fire Safety Journal

journal homepage: www.elsevier.com/locate/firesaf

Evolution of combustion gas concentrations in full-scale residential fire environments

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ARTICLE INFO

Keywords:
Compartment fires
Residential fires
Gas transport
Toxic gas exposure
Hydrogen cyanide
Tunable diode laser
Absorption spectroscopy

ABSTRACT

Limited availability of time-resolved hydrogen cyanide (HCN) concentration in full-scale residential fires curtails the ability to estimate toxic smoke exposure to a potential occupant. Sixteen experiments simulating bedroom and kitchen fire scenarios with realistic fuel loads were conducted in a full-scale residential structure. Trends in time-resolved concentration of oxygen (O_2), carbon dioxide (CO_2), carbon monoxide (CO) and HCN were analyzed. The toxic gas exposure potential of a central location in the structure, prior to firefighter intervention, was evaluated. Fire grew more rapidly in bedroom experiments. Concentration of measured gases changed post-flashover in bedroom experiments and pre-flashover in kitchen experiments. The average rates of change for O_2, CO_2 and CO were 3 times higher in the bedroom experiments and with a narrower range compared to kitchen experiments but were similar for HCN in both scenarios. Peak concentrations were higher in kitchen experiments with immediately dangerous to life and health (IDLH) condition first caused by low O_2 followed by high HCN and CO. Time to IDLH was 3 times shorter in bedroom experiments. Based on toxic gas fractional effective dose, the sampling location remained tenable for about 3 times longer in kitchen experiments despite the larger fire size and greater peak concentrations.

1. Introduction

In a residential fire scenario, exposure to excessive heat and smoke are the two main factors that make the residential fire environment untenable for a potential trapped occupant [1,2]. The effects of high temperature on a potential trapped occupant can take the form of skin burns and respiratory burns of varying severity, based on their location relative to the fire and the exposure time [3,4]. However, the effects of smoke exposure are notably more complex. Particulates in the smoke reduce visibility, which can slow egress from the structure and increase exposure time. Some of these substances have a short term effect upon exposure that are reversible once exposure ceases. For example, exposure to carbon dioxide (CO_2) causes hyperventilation and exposure to acid gases cause irritation of the skin, eyes and respiratory tract. Some substances in the smoke may be acutely fatal or have a lasting impact on the health of the exposed person depending on the quantity and duration of exposure [5–7]. Asphyxiant gases such as carbon monoxide (CO) and hydrogen cyanide (HCN), are of particular interest as they are not only acutely toxic in relatively small concentrations, but are also known to have a synergistic effect and exacerbate the toxic effects of smoke exposure [8–10]. Moreover, the severity of the hazard depends on several factors such as distance from the fire compartment, quantity and composition of the fuel, availability of oxygen to the fire, layout of structure.

After CO is inhaled, it is introduced into the blood stream, where it combines with hemoglobin to form carboxyhaemoglobin (COHb). COHb remains in the blood for several days after an exposure and therefore can be measured at most hospitals and medical facilities to estimate the extent of exposure. For this reason, fatalities and injuries in fires due to CO poisoning are well documented [11]. However, cyanide metabolizes rapidly after exposure and blood samples are typically not collected and analyzed soon enough after fire incidents to provide an accurate measure of HCN exposure [9,12,13]. Recently, a prototype field-portable sensor for rapid analysis of blood cyanide has been developed [14]. However, making such sensors widely available may not be immediately practical. Full-scale studies evaluating the smoke toxicity and fire gas composition in compartment fires have provided some insight into the effects of smoke exposure, especially CO and HCN, on potential trapped occupants. Experiments involving ventilation-limited fires, have been conducted in multi-compartment structures with single component fuels as well as realistic residential fuel loads [15–20]. However, there is limited availability of time-resolved HCN concentration data along with the other combustion gases. This has led to researchers utilizing bench-scale experiments data to predict gas transport and estimate smoke toxicity for full-scale fire scenarios by applying a scaling factor [21–24]. However, the scalability

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https://doi.org/10.1016/j.firesaf.2026.104724
Received 19 November 2025; Received in revised form 28 January 2026; Accepted 11 March 2026
Available online 17 March 2026

of bench-scale data to full-scale scenarios cannot be sufficiently verified without full-scale smoke composition datasets.

The aim of this work is to characterize the cumulative influence of fuel composition, ignition location and external ventilation changes on the concentration of combustion products, including CO and HCN, at a central location in a residential structure as the fire is ignited and transitions through flashover prior to firefighter intervention. The purpose of this paper is to add to the available literature on full-scale experimental data by analyzing the trends in time-resolved concentration of HCN along with O_2, CO_2 and CO and evaluate the toxic gas exposure potential of the location for bedroom and kitchen fire scenarios with realistic fuel load.

2. Methods

A series of full-scale experiments was conducted in two identical 160 m^2 ranch-style single storey residential structures fully furnished with a realistic residential fuel load. The overall objective of these experiments was to study the impact of the ignition location, the isolation of fire and non-fire compartments using closed doors, and search and rescue tactics, on firefighter safety and occupant survivability. A detailed description of the experiment methods and the finding of the larger study are detailed in technical reports published by Fire Safety Research Institute [25,26]. A summary of the relevant parameters is provided in the following sub-sections. The experiments included in the current study are tabulated in Table 1. For consistency with the technical reports, the experiments are referred to by the same labels in this paper.

2.1. Experimental setup

Experiments were conducted at the Delaware County Emergency Services Training Center, Sharon Hill, PA. The experiment structures featured four bedrooms, two bathrooms and an open concept kitchen and living room as shown in Fig. 1, with the ceiling height of 2.45 m. The walls were comprised of 38 mm × 90 mm No. 2 grade SPF (Spruce-Pine-Fir) wood studs filled with R-13 fiberglass insulation, and lined with 15.8 mm Type X gypsum wallboard and two coats of white latex flat wall paint on the interior. On the exterior, the walls were sheathed with 1.11 cm oriented strand board, olefin home wrap, and 6.35 mm fiber cement board siding. All windows in the structure were dual pane, double-hung windows except for the bathroom windows which were dual pane and non-operable. Hollow-core wood frame doors were used as internal doors to bedrooms, closets, and bathrooms. A closed-loop heating, ventilation, and air-conditioning (HVAC) system was installed in the structure. A total of 9 supply vents, one in each of the bedrooms and bathrooms, one in the kitchen and two in the living room, were located on the ceiling and measured 35 cm × 17 cm. The return vents, measuring 35 cm × 20 cm were located 20 cm above the floor, and situated in each of the bedrooms, in the living room and in the hallway. The HVAC system was off, but all of the vents were open to allow for passive air and gas movement within the structure.

The experiment structure was furnished to represent a realistic residence as shown in Fig. 2. In the bedroom compartment, wood-based fuel made up about 28 % of the available fuel by weight and weighed about 67 kg. The non-wood fuel weighed about 168 kg. In contrast, about 53 % of the fuel in the kitchen-living room compartment was wood-based and weighed about 489 kg. Non-wood based fuel in the kitchen-living room compartment weighed about 435 kg. The non-wood fuel throughout structure was comprised primarily of synthetic polymers such as polyethylene, polyethylene terephthalate, polyurethane and polyester.

As indicated in Fig. 1, the fire was either ignited on the bed-side chair in bedroom 4 for bedroom experiments or on the kitchen counter for kitchen experiments. For bedroom experiments, the fire was ignited remotely using an electric match placed on the rear cushion of the bed-side chair with a small amount of the polyester batting pulled out to facilitate ignition as shown in Fig. 2(d). For the kitchen experiments, a 4 kW propane burner was used to heat 180 mL of canola oil to its auto-ignition temperature in an aluminum pan to simulate an unattended cooking fire scenario. To expedite the initial fire growth, a paper towel roll, bags of chips, and disposable cups made of paper and plastic are placed near the burner-pan setup as shown in Fig. 2(e). For a complete description of the fuel, the reader is referred to the technical reports [25,26].

All windows except the fire compartment window started in the closed position. For all the bedroom experiments, the bottom sash of the bedroom 4 window was removed (1.8 m × 0.6 m, 0.9 m sill height). Likewise, the kitchen window was removed for the kitchen experiments (0.9 m × 0.9 m, 1.2 m sill height). The front door was open in all experiments providing the other initial source of external ventilation (0.9 m × 0.9 m). As for the interior doors, bedroom 1 door started in the closed position (expect for Experiment 17), bedroom 2 door started in the open position and bedroom 3 door started in the open position (except for Experiment 8b). Bedroom 4 door started in the closed position for some of the kitchen experiments (Experiments 11, 12, 13 and 18). For all other experiments, the bedroom 4 door was either removed or left open.

The fire compartment transitioned through flashover before any firefighter intervention activities were conducted. Flashover was determined visually by monitoring video cameras placed in the fire compartment in conjunction with the temperature readings from the vertical array of 8 bare-bead K-type thermocouples with bead diameter of 1.27 mm placed in that room. The locations on the thermocouple arrays are indicated by green triangles in Fig. 1. Post-flashover, various search tactics were simulated either before or during suppression. The fire suppression crew entered the structure through the front door and extinguished the fire using water. After an experiment was concluded, the structure was overhauled by replacing damaged doors, drywall, flooring, windows, etc. It was then re-painted, re-furnished and re-instrumented to minimize the effects of repeated fire damage to the structure.

The current work focuses on the fire growth from ignition to the first firefighter intervention activity in a subset of 16 experiments from the larger study. Time-resolved concentrations of O_2, CO_2, CO and HCN, were measured in the gas sampled from the end of the hallway (indicated by the blue circle in Fig. 1) at 0.9 m above the floor are examined. The 0.9 m elevation was chosen to represent the approximate kneeling or crawling height of a potential trapped occupant. This location is suitable for this study as it is least likely to be influenced by outdoor weather conditions and therefore could be expected to respond in a similar fashion for experiments conducted on different days. Moreover, of all the locations at which gas concentrations were measured in the larger study, the gas concentrations at the end of the hallway were the first to change [25,26]. Although this location is not in the direct flow path between the fire compartment and the external vents, the gas flow stagnates at the wall, suggesting that toxic gas exposure might be a greater hazard than thermal exposure at this location.

2.2. Measurement systems

The gas mixture was sampled continuously using a diaphragm pump that pulled the gas mixture through a stainless steel intake port. It was then filtered through 5 μm and 3 μm HEPA filters to remove particulate matter and passed through a condenser (stainless steel coil in an ice-water bath) to remove moisture. The sampled gas was split into two sample lines, one of which was passed over a desiccant column filled with Drierite to ensure complete elimination of moisture before being introduced into a Siemens ULTRAMAT-23 analyzer. CO_2 and CO were measured by a non-dispersive infrared (NDIR) sensor and O_2 was measured by a paramagnetic sensor. The CO analyzer has a range of

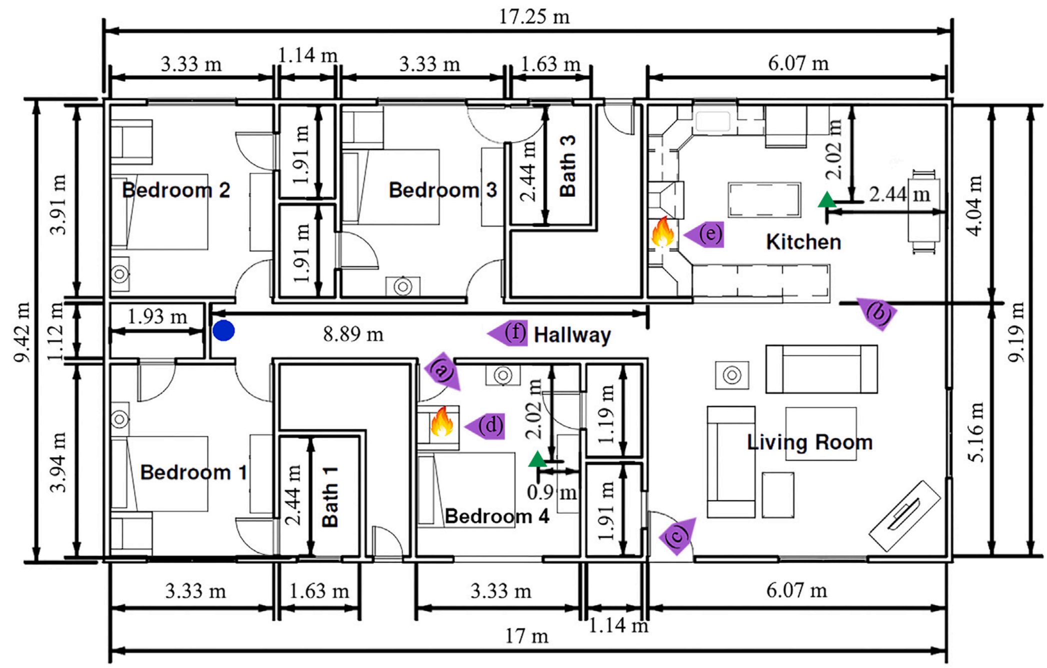
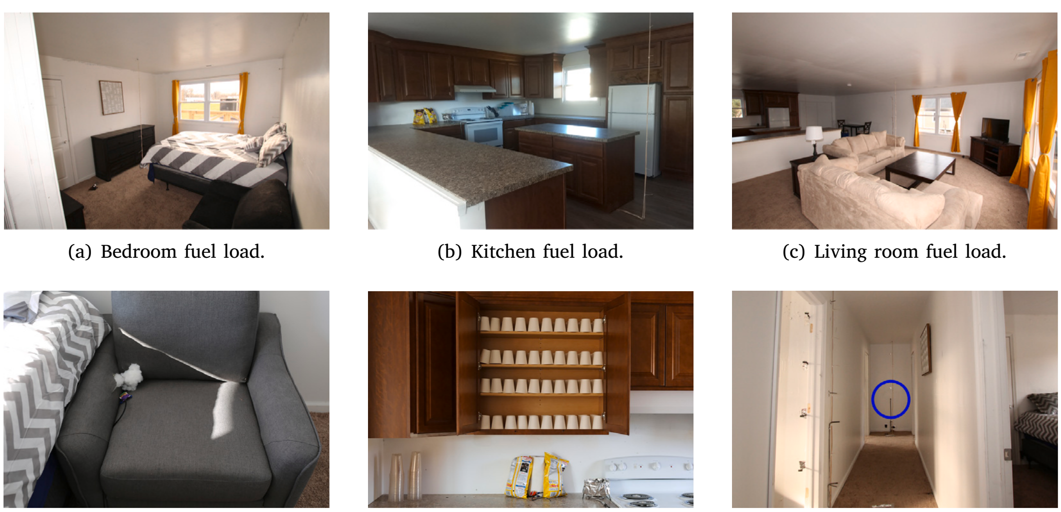


Fig. 1. Dimensioned layout of the experiment structure. The measurement location is indicated by a blue circle, and the ignition locations for both the bedroom and kitchen experiments are indicated by the flame symbols. (Note that there was a single ignition source per experiment.) The green triangles indicate thermocouple arrays in the fire compartments. The point of view for representative images shown in Fig. 2 are indicated by labeled purple flags. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)



(a) Bedroom fuel load. (b) Kitchen fuel load. (c) Living room fuel load.

(d) Bedroom ignition setup. (e) Kitchen ignition setup. (f) Sampling location.

Fig. 2. Images representing the fuel load and ignition setups used for the experiments along with the sampling location indicated by the blue circle. The locations and points of view for the images are indicated by labeled purple flags on the layout (Fig. 1). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

0–5 vol% and a resolution of 0.001 vol%. The CO_2 and O_2 analyzers have a range of 0–25 vol% and a resolution of 0.01 vol%. Similar gas sampling instruments have demonstrated a relative expanded uncertainty of ± 1 % when compared to span gas volume fractions [27]. The other sample line was introduced into a mid-infrared tunable diode laser absorption spectroscopy (TDLAS) based HCN measurement system which has a limit of detection of 1 ppm and a limit of quantitation of 3 ppm [28]. The relative uncertainty for measurement of HCN is 4.1 % at 1500 ppm. The sampling frequency was 1 Hz for all measurements. Fig. 3 shows a representative time resolved concentrations of the species measured in the gas sampled from the end of the hallway at 0.9 m height for Experiment 6. Some of the notable events for the experiment are marked using vertical dotted lines.

Gas transport time from the sampling location was measured prior to each experiment and ranged from 16 to 23 s. The data acquisition system was used to measure the time to change in concentration values on the gas analyzers from the start of a 3–5 s discharge of calibration gas across the sampling port. The two calibration gases used were composed of 2.51 vol% CO, 12.60 vol% CO_2, balance N_2, and 50 ppm HCN, balance N_2. The uncertainty in measuring transport time was determined to be at least 3.4 s. However, for this study, a conservative approach is adopted and gas concentration changes less than 5 s (0.08 min) apart are considered simultaneous responses.

2.3. Analysis methods

At least 2 min of background data was recorded at the start of each experiment. In the case of bedroom experiments, the time at which the electric match was triggered was marked as the ignition. For kitchen experiments, the time of auto-ignition of the cooking oil was designated as the ignition. All the subsequent events in the experiment were recorded relative to ignition. The data is cutoff at the time of the first intervention activity, henceforth referred to as intervention, that may have an influence on the fire such as opening a door or a window to enter the structure or water application. The time of flashover of the fire compartment is identified as the time when the temperature at 0.9 m elevation in the fire compartment exceeds 600 °C [29–31]. As fire transitioned through flashover, external vent failure was observed in the fire compartment (i.e. breakage of glass in the top sash of bedroom 4 window for bedroom experiments and breakage of the glass in the living room windows for kitchen experiments). The time at which the window glass broke and flames were visible outside the structure is designated as the time of vent failure (marked as 'Fire room vent failure' in Fig. 3).

The change in concentration of the combustion gases was calculated as an absolute difference between the measured concentration and the average background concentration. A linear baseline was inferred from the change in concentration data, from the start of background to time of intervention, using an iterative polynomial fit algorithm, The last time index, prior to intervention, at which the change in gas concentration intersects with its baseline was determined to be the time of initial response in the gas concentration. The time of initial response in O_2 concentration ($t_{\Delta O_2}$) at the sampling location is indicative of the ambient air being displaced by the combustion products transported from the fire compartment. The average rate of change in gas concentration was determined by averaging the instantaneous rate of change from $t_{\Delta O_2}$ until intervention. The maximum change in concentration of the measured gases was also calculated. Based on the ignition location, average values and their sample standard deviation from the mean were calculated and compared.

To assess the extent of the hazard of asphyxiant gas exposure at the sampling location, the Purser model [11] is used to estimate the fractional effective dose (FED) from the measured time-resolved concentrations of O_2, CO_2, CO and HCN. FED = 1 represents a dose predicted to cause collapse and loss of consciousness due to asphyxia in 50 % of the population. To accommodate vulnerable sub-populations, ISO13571 recommends a more conservative limit of FED = 0.3 for general population [32]. FED = 3 represents a dose predicted to incapacitate even the healthiest of the population [33]. The time from ignition to reach these FED thresholds is calculated.

3. Results and discussion

For the analysis presented in this paper, 10 bedroom experiments and 6 kitchen experiments are used. Unless noted, errors mentioned here are estimated as expanded uncertainty (coverage factor of 1) calculated as sample standard deviation for respective set of bedroom or kitchen experiments for various relevant parameters. The experiments are listed in Table 1, along with the ignition location, and the times of flashover, external vent failure of the fire compartment, and intervention, relative to the time of ignition. The average time to flashover for kitchen experiments was approximately 12 min greater than the average time to flashover for bedroom experiments. This could be attributed to the kitchen-living room compartment being about 5.4 times larger by volume than the bedroom compartment and the composition of the fuel involved during initial fire growth. The predominantly synthetic polymer based fuel package in the bedroom compartment

Table 1
Summary of times relative to ignition for pertinent fire behavior events.

Ignition location	Experiment number	Flashover (min)	Vent failure (min)	Intervention (min)
Bedroom 4	2	3.5	4.3	5.3
	3	3.3	4.0	5.0
	4	3.1	4.6	4.5
	5	3.2	3.6	5.0
	6	3.4	4.1	5.3
	7	2.8	3.3	5.3
	8	2.9	3.3	4.8
	8b	3.1	3.7	4.5
	9	2.7	3.4	4.8
	10	3.2	3.4	5.0
Avg. (Std. Dev.)		3.1 (0.3)	3.8 (0.4)	4.9 (0.3)
Kitchen	11	15.0	16.1	16.2
	12	14.9	15.8	17.0
	13	15.0	15.6	15.5
	14	14.1	16.1	17.1
	17	14.7	16.1	16.9
	18	15.8	11.5	18.1
Avg. (Std. Dev.)		14.9 (0.5)	15.2 (1.8)	16.8 (0.9)

compared to a mostly wood based fuel package in the kitchen-living room compartment most likely, in addition to the impact of compartment volumes, resulted in the bedroom experiments progressing more rapidly compared to the kitchen experiments. Additionally, the compact arrangement of furniture in the bedroom compartment likely promoted rapid spread of the fire and quicker transition to compartment flashover. Failure of the bedroom window occurred on average 0.7 min after flashover of the bedroom compartment, whereas it took 1.2 min on average, excluding Experiment 18, for the living room windows to fail after flashover of the kitchen-living room compartment. Although Experiment 18 followed the general trend for timing of fire growth, the living room window for this experiment failed prior to flashover of the kitchen-living room compartment.

3.1. Time of initial response

Table 2 lists the time of initial response in gas concentration with the average time to change in concentration values and the standard deviation calculated for bedroom and kitchen experiments. At the end of the hallway at 0.9 m elevation, O_2 concentration started to decrease from the ambient, on average, 3.3 ± 0.3 min post-ignition for bedroom experiments. Increases in concentration of CO_2, CO and HCN were observed around the same time the O_2 concentration began to decrease, with a similarly low variability between experiments. The fire in the bedroom experiments progressed from clean burning to under-ventilated through flashover in about 3 min after ignition. This likely resulted in initial responses in gas concentrations at the sampling location occurring almost simultaneously. For the kitchen experiments, the fire progressed much slower, with O_2 and CO_2 concentrations changing first, indicating a clean burn. The time to change in O_2 and CO_2 concentrations, relative to ignition, was about 2 times that for bedroom experiments, with greater variability between experiments as evident from their sample standard deviation of 1.4 min and 1.6 min respectively. As the fire progressed to under-ventilated conditions, changes in concentrations of CO and HCN were also observed. For all kitchen experiments, except Experiment 18, the CO concentration began to increase, on average 1.4 min after an increase in CO_2 concentration was observed which was followed by an increase in HCN concentration, on average 2.8 min after CO concentration increase. This difference in time to change in gas concentration based on ignition location was likely a result of the relative proximity of the sampling location to the fire for bedroom experiments compared to kitchen experiments in addition to the different rates of fire progression in the two scenarios. The patterns of flow of fresh air and combustion gases

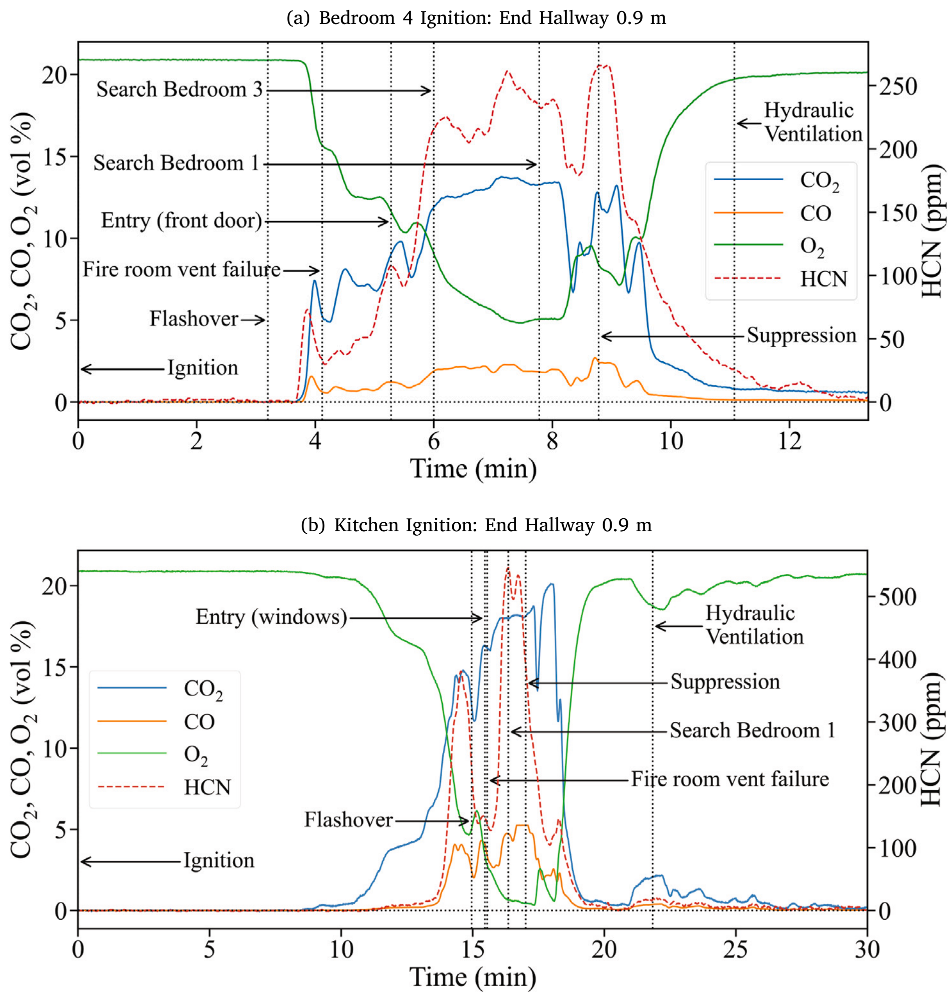


Fig. 3. Representative datasets showing the time evolution of oxygen and the combustion gases at the end of the hallway at 0.9 m height. The first firefighter intervention activity for the Bedroom 4 experiment was ‘Entry (front door)’, and for the Kitchen experiment was ‘Entry (windows)’.

established due to the placement of the sampling location relative to the fuel load as well as the external ventilation configuration likely influenced the times to change in gas concentration. For the bedroom experiments, the combustion products were observed to fill the hallway before spreading to the rest of the structure, whereas, for the kitchen experiments, the combustion products were observed to spread in the kitchen-living room compartment before spreading to the 0.9 m height sampling location in the hallway. A study conducted by Forrest et al. in a 2-storey structure with typical living room furnishings, shows similar pattern in gas evolution at the 0.9 m height where CO and HCN are observed to increase 1–2 min after change in O_2 concentration was detected [34]. Experiment 18 deviated from the pattern of the other kitchen experiments as the HCN concentration was observed to increase about 3 min after ignition, which was followed by change in CO, CO_2 and O_2 concentrations about 3.2 min later.

Comparing the timing of fire behavior events and time of change in concentration, for bedroom experiments, the O_2 concentration began to change on average 0.2 min after fire compartment flashover but 0.5 min before failure of the bedroom 4 window. However, for the kitchen experiments, the O_2 concentration in the sampled gas began to change on average 8.2 min before the kitchen-living room compartment flashover. Living room window failure occurred after fire compartment flashover for all kitchen experiments except Experiment 18.

Table 2
Time of initial response in gas concentration at the end of the hallway at 0.9 m elevation relative to ignition.

Experiment number	Response in O_2 (min)	Response in CO_2 (min)	Response in CO (min)	Response in HCN (min)
2	3.8	3.7	3.8	3.4
3	3.6	3.6	3.7	3.1
4	3.2	3.3	3.3	3.2
5	3.5	3.4	3.4	3.4
6	3.5	3.7	3.7	3.7
7	3.0	3.0	3.0	3.1
8	2.9	3.0	3.2	3.1
8b	3.3	3.3	3.4	3.7
9	2.9	2.9	3.2	3.1
10	3.2	3.3	3.4	3.4
Bedroom 4 Avg (Std. Dev.)	3.3 (0.3)	3.3 (0.3)	3.4 (0.3)	3.3 (0.2)
11	4.2	4.1	4.8	9.3
12	6.3	6.3	7.6	12.4
13	8.3	8.6	9.1	11.2
14	7.8	7.9	8.9	10.7
17	7.5	7.4	11.1	12.2
18	6.3	6.4	6.3	3.1
Kitchen Avg (Std. Dev.)	6.7 (1.4)	6.8 (1.6)	8.0 (2.2)	9.8 (3.5)

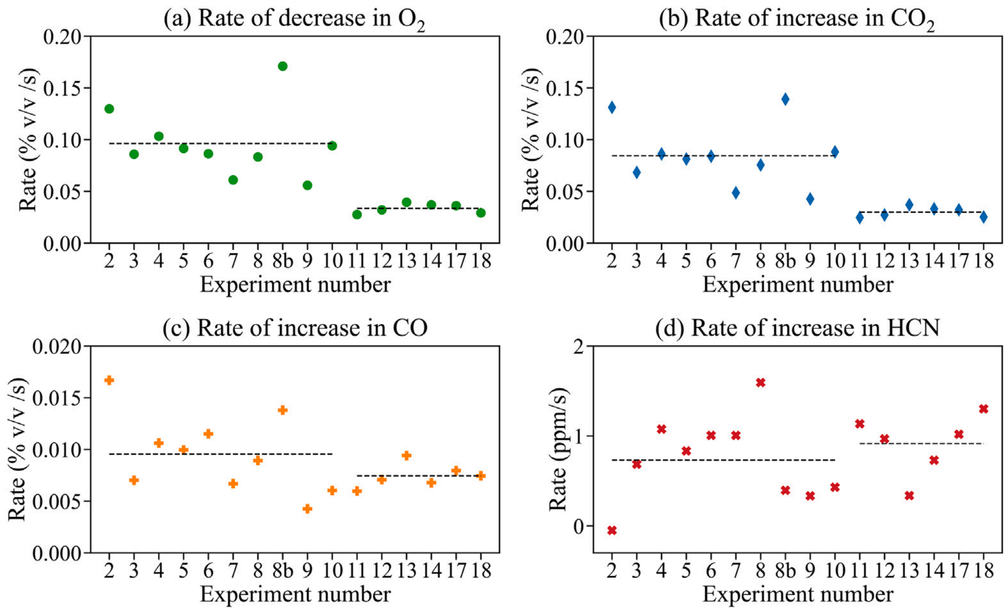


Fig. 4. Average of instantaneous rate of change in gas concentrations from time of initial change in O_2 concentration until intervention.

3.2. Rate of change

The average rate of change in gas concentration of O_2, CO_2, CO and HCN from $t_{\Delta O_2}$ to intervention is plotted in Fig. 4. Prior to $t_{\Delta O_2}$, the average rate of change for all the experiments was 0 vol%/s for O_2, CO_2 and CO concentrations, and 0 ppm/s for HCN concentration for all experiments. The average rate of change in the concentration of measured gases, while making it easier to compare between the two ignition scenarios, was not representative of the range of rates of change observed in the gas concentration data. The composition and placement of fuel in the fire compartment resulted in different rates of production of combustion gases from multiple sources in different stages of combustion. In contrast, for a series of experiments conducted in a different multi-compartment structure, using propane gas as fuel, the rate of change in concentration after $t_{\Delta O_2}$ till flame extinction due to fuel shut-off, was observed to be constant [35].

After $t_{\Delta O_2}$, the rate of decrease in O_2 concentration ranged from −0.11 vol%/s to 0.61 vol%/s and averaged 0.10 vol%/s until intervention for all bedroom experiments. For kitchen experiments, the rate of decrease in O_2 concentration ranged from −0.24 vol%/s to 0.72 vol%/s and averaged 0.03 vol%/s. For CO_2 concentration, the rate of increase ranged from −0.402 to 1.06 vol%/s, averaging 0.09 vol%/s over the bedroom experiments, and ranged from −0.66 vol%/s to 1.15 vol%/s, averaging 0.03 vol%/s over the kitchen experiments. The rate of increase in CO concentration ranged from −0.22 vol%/s to 0.34 vol%/s, averaging 0.01 vol%/s over the bedroom experiments, and ranged from −0.29 vol%/s to 0.39 vol%/s, averaging 0.008 vol%/s over the kitchen experiments. For HCN concentration, the rate of increase ranged from −9.3 ppm/s to 15.3 ppm/s, averaging 0.7 ppm/s over bedroom experiments, and ranged from −24.5 ppm/s to 34.3 ppm/s, averaging 0.9 ppm/s over kitchen experiments.

In the kitchen experiments, the average rate of decrease in O_2 concentration was about 3.3 times lower, and the average rate of increase in CO_2 and CO concentrations was about 3 times and 1.25 times lower, respectively, compared to the bedroom experiments. In contrast, the average rate of increase in HCN concentration was about 1.3 times higher in kitchen experiments compared to bedroom experiments. The lower average rate of change in concentration at the sampling location observed in kitchen experiments could be attributed to the location of open front door relative to the fire and the measurement location. As shown in Fig. 1, the front door, an efficient vent for gas exchange, was between the fire and the end of the hallway. For the bedroom experiments, combustion products that were transported from the interior bedroom 4 door, accumulated in the hallway with limited pathways for mixing with fresh air from an exterior vent.

3.3. Maximum change

Fig. 5 shows the maximum change in concentration of O_2, CO_2, CO and HCN relative to their respective background concentration. The average change in CO_2, CO and HCN concentration before $t_{\Delta O_2}$ for bedroom experiments was 0.06 ± 0.01 vol%, 0.01 vol% and 9 ± 13 ppm respectively. For kitchen experiments, the change in CO_2, CO and HCN concentration before $t_{\Delta O_2}$ averaged to 0.09 ± 0.04 vol%, 0.01 vol% and 7 ± 3 ppm respectively. After $t_{\Delta O_2}$, the maximum decrease in O_2 concentration prior to intervention averaged to 9.25 ± 1.65 vol% in bedroom experiments. The maximum increase in the CO_2, CO and HCN concentration was, on average, 8.53 ± 1.85 vol%, 1.42 ± 0.50 vol% and 97 ± 47 ppm respectively, across all bedroom experiments. For kitchen experiments, the maximum decrease in O_2 concentration averaged to 19.88 ± 1.27 vol%. The maximum increase in the CO_2 and HCN concentration was, on average, 19.79 ± 0.77 vol% and 660 ± 211 ppm respectively. For Experiments 11, 12 and 18, the CO sensor for the sampling location reached its upper measurement limit of 5 vol% CO by volume. However, the average maximum increase in CO concentration for kitchen experiments can be estimated to be greater than 4.83 ± 0.46 vol%.

Prior to intervention, the average maximum change in O_2 and CO_2 concentration was 2.2 times and 2.1 times greater, respectively, for kitchen experiments compared to bedroom experiments. The average maximum change in CO concentration was at least 3.4 times greater and that in HCN concentration was 6.8 times greater for kitchen experiments compared to bedroom experiments. This was most likely due to availability of greater quantity of fuel as well as more ventilation in the kitchen-living room area resulting in larger fire size in the kitchen experiments. The initial fuel load in the kitchen-living room compartment was about 923 kg compared to the initial fuel load of about 234 kg in the bedroom compartment. The initial external ventilation for the bedroom experiments consisted of open bedroom window and open front door, the combined area of the openings measuring about 2.97 m^2. However, as the fire transitioned to flashover, the failure of the bedroom window increased the external ventilation area to

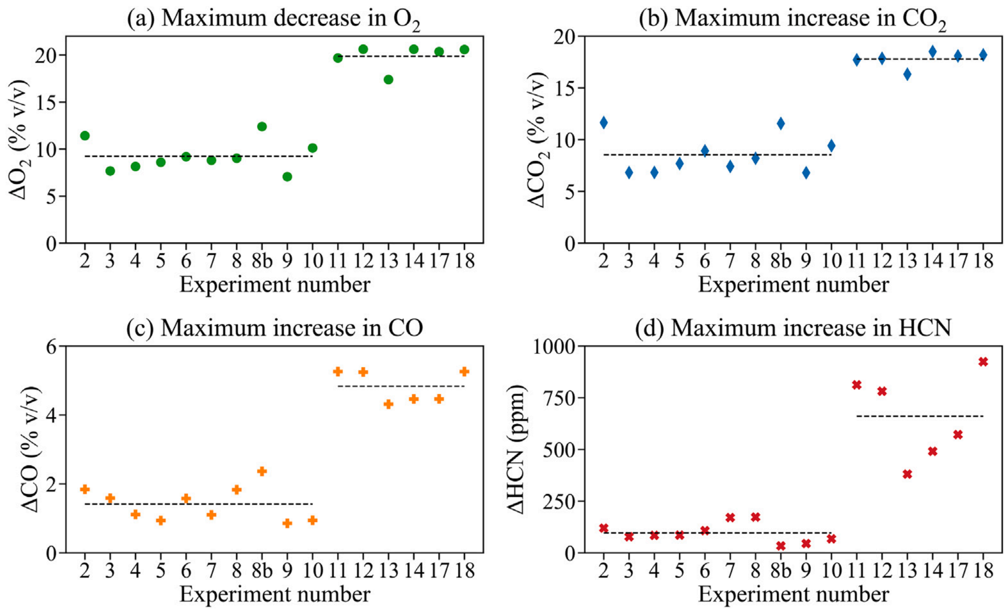


Fig. 5. Maximum change in gas concentrations from time of initial change in O_2 concentration until intervention.

about 4.09 m^2. The kitchen experiments had similar initial ventilation conditions, open kitchen window and open front door, with a combined area of the openings measuring about 2.28 m^2. As the kitchen fire transitioned through flashover, the kitchen window failed, increasing ventilation area to about 2.69 m^2. As the fire spread to the living room, the flashover of the kitchen-living room compartment resulted in failure of both the living room windows, providing an additional increase in ventilation to about 8.27 m^2. The post-flashover ventilation for kitchen experiments was 2 times larger than bedroom experiments however the fuel quantity in the kitchen experiments was 3.94 times greater than bedroom experiments. This could be a key reason for the greater increase in concentration of products of incomplete combustion, CO and HCN, observed for kitchen experiments compared to bedroom experiments.

3.4. Time to compromised tenability

According to the National Institute for Occupational Safety and Health (NIOSH), the immediately dangerous to life and health (IDLH) exposure concentration for CO_2 is 40,000 ppm (40 vol%), for CO is 1200 ppm (1.2 vol%) and for HCN is 50 ppm [36]. Oxygen concentration of less than 19.5 vol% is considered IDLH [37]. Therefore, in context of data presented in this work, decrease in O_2 concentration greater than 1.4 vol% would be considered IDLH condition. Meeting any one of these conditions would qualify the sampling location to be in IDLH condition.

For all the experiments, regardless of the ignition location, the IDLH condition due to toxic gas exposure was reached at the 0.9 m elevation at the end of the hallway prior to intervention, either due to low O_2 concentration or concentration of CO or HCN exceeding their respective IDLH levels. Concentration of CO_2 remained below IDLH level prior to intervention for all experiments. For the six kitchen experiments, intervention occurred approximately 17 min after ignition. The IDLH condition was met due to low O_2 concentration 6.1 ± 2.0 min before intervention which was followed by HCN and CO exceeding their respective IDLH concentrations, 2.3 ± 0.7 min and 2.2 ± 0.7 min before intervention. For the ten bedroom experiments, intervention occurred 4.9 ± 0.3 min after ignition. For four of the bedroom experiments (Experiments 2, 3, 6 and 8), the IDLH condition was reached due to HCN exceeding IDLH concentration 1.4 ± 0.2 min before intervention, followed by CO and O_2 exceeding IDLH levels 1.3 ± 0.2 min and 1.2 ± 0.2 min before intervention, respectively. For another set of four experiments (Experiments 4, 5, 7 and 10), the IDLH condition was reached due to low O_2 concentration, 1.4 ± 0.4 min before intervention, followed by HCN exceeding IDLH concentration, 0.8 ± 0.6 min before intervention. Concentration of CO remained below IDLH value prior to intervention for this set of experiments. For Experiment 8b, IDLH was reached due to low O_2 concentration, 0.8 min before intervention, followed by CO exceeding IDLH concentration, 0.7 min before intervention. For Experiment 9, IDLH condition was reached due to low O_2 concentration, 1.3 min prior to intervention while the other species of gases measured remained within their respective IDLH concentration thresholds. The times at which O_2, CO and HCN concentration cross their respective IDLH levels along with the time of intervention relative to ignition are plotted in Fig. 6. The time to IDLH condition calculated based on individual gas concentration was 3.6 ± 0.2 min after ignition for bedroom experiments and 10.7 ± 1.7 min after ignition for kitchen experiments.

The cumulative effect of the concentrations of measured gases, calculated based on fractional effective dose, results in time to compromised tenability that is slightly shorter for the kitchen scenario, and comparable to time to IDLH in the bedroom fire scenario. Also plotted in Fig. 6 are the times at which FED thresholds for incapacitation for general population ($FED_{IN} = 0.3$), for 50 % of the healthy adults ($FED_{IN} = 1$) and for the healthiest of the population ($FED_{IN} = 3$) are exceeded. For bedroom experiments, the average time to $FED_{IN} = 0.3$ and $FED_{IN} = 1$ were both calculated to be 3.7 ± 0.2 min. For kitchen experiments, the average time to $FED_{IN} = 0.3$ was calculated to be 9.1 ± 2.0 min and average time to $FED_{IN} = 1$ was 10.4 ± 2.1 min. The time to $FED_{IN} = 3$ was on average 0.1 ± 0.1 min longer than the time to $FED_{IN} = 0.3$ in bedroom experiments, and 2.0 ± 0.5 min longer than the time to $FED_{IN} = 0.3$ in kitchen experiments.

4. Limitations and future work

Although there are clear differences in the evolution of gas concentrations between the bedroom and kitchen fire scenarios, the layout and ventilation conditions are unique to these experiments. Additional studies with different layouts, comparable fuel load and gas sampling location may enable generalization of these trends to all structures. As these experiments were conducted outdoors, the influence of external weather conditions such as wind speed and direction, temperature,

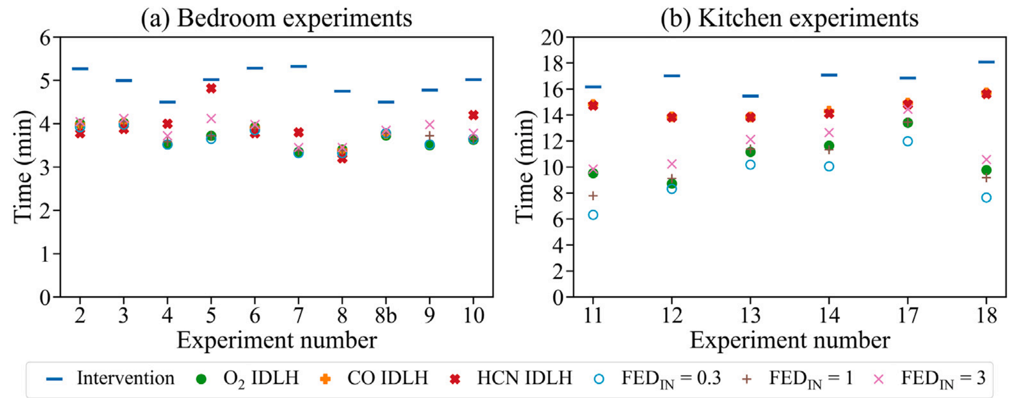


Fig. 6. Time of intervention and time at which measured gas concentrations cross IDLH and FED thresholds due to asphyxiant gas exposure relative to ignition.

humidity, etc. could not be decoupled from the fire-driven transport of gases inside the structure. Full-scale experiments inside a fire lab in the future may better isolate the experiment structure from the influence of external weather conditions. Moreover, the trends in gas concentrations reported in this work are due to the combined effect of gases produced by the fire and gas transport driven by the fire. A fire modeling study could be performed in the future to estimate gas concentrations at the sampling location using small-scale gas yields for representatives of the combustibles used in these experiments.

The gas sampling technique used in these experiments removed the moisture content without quantifying it. Consequently, the measured gas concentration data can only be used to quantify and compare the trends to be expected at the sampling location based on the dry-gas concentrations. Measuring time-resolved concentration of water vapor in the sampled gas may give a more accurate measure of the gas composition at the sampling location inside the structure for future studies.

5. Summary and conclusion

Time-resolved concentrations of gases sampled at a kneeling or crawling height (0.9 m height above the floor) of a potentially trapped occupant in a representative compartment fire scenario were analyzed for a total of 16 full-scale residential fire experiments and clear distinctions are evident in gas concentration data for bedroom and kitchen fire scenarios. Based on the fire compartment volume, available fuel and ventilation, the fire size in bedroom experiments was smaller compared to kitchen experiments. On average, kitchen experiments took about 4.8 times longer to flashover the fire compartment and about 1.8 times longer for external vent failure post-flashover compared to bedroom experiments. Concentrations of the measured gases began to change around the same time for the bedroom experiments after flashover but before external vent failure with low variability between experiments. For kitchen experiments, concentrations of all the measured gases began to change prior to flashover. Concentrations of O_2 and CO_2 took about 2 times longer to change when compared to bedroom experiments and were followed by change in CO concentration and then in HCN concentration few minutes later. The average rates of change in concentration of O_2, CO_2, and CO for bedroom experiments were about 3 times higher and with a wider range when compared to kitchen experiments while the average rate of change in HCN concentration was similar for bedroom and kitchen experiments. Despite having lower rates of change, kitchen experiments were observed to have greater maximum change in concentration prior to intervention due to the longer duration of kitchen experiments. The average maximum change in the concentrations of O_2 was about 2.2 times, CO_2 was about 2.1 times, CO was about 3.4 times and HCN was about 6.8 times higher than that observed for bedroom experiments. The IDLH condition was first reached due to low O_2 concentration for kitchen experiments, followed by HCN and CO exceeding their respective IDLH concentration. For the bedroom experiments no gas could be singled out as the first cause of IDLH condition. Time to IDLH condition post-ignition was about 3 times shorter for bedroom experiments compared to kitchen experiments. Considering the cumulative effects of the measured gases based on FED for toxic gas exposure, the sampling location remained tenable for about 3 times longer for kitchen experiments compared to bedroom experiments.

The analysis of time-resolved concentration of gases sampled from a central location in the residential structure provided an insight into the gas transport to and toxic gas exposure potential of that location. Moreover, it is evident that although kitchen experiments had larger fires and generated more amount of toxic gases, the bedroom experiments were faster and therefore more dangerous due to toxic gas exposure to potential occupants prior to firefighter intervention.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The author would like to thank Gavin Horn, Nick Dow and Craig Weinschenk for their support during the development of this manuscript.

The experiments were supported by Grant EMW-2017-FP-00628 from the United States Department of Homeland Security (DHS) Federal Emergency Management Agency's Assistance to Firefighters Grant program under the Fire Prevention and Safety Grants: Research and Development.

Data availability

Data will be made available on request.

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